

Commitment To Cooperation With Self-Negotiated Contracts

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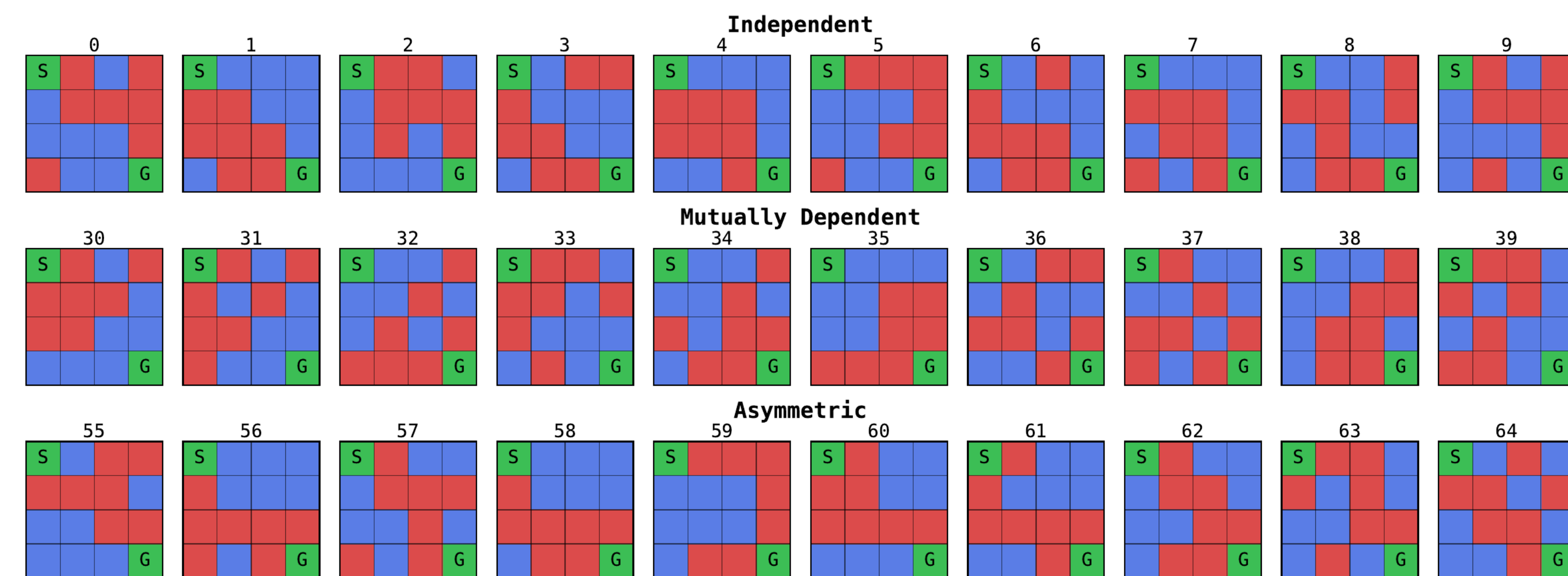
Why Contracts?

Cooperation is hard: its **costs come early**, its **benefits come later** — an incentive to defect. Human societies solve this with **contracts**: observable agreements whose enforced terms make commitments credible.

Can LLM agents use self-negotiated contracts to cooperate with commitment?

Board Types (80 boards)

- Independent (20)**: each player can finish alone
- Mutually Dependent (20)**: neither can finish without trading
- Asymmetric (40)**: **P-Red** can finish alone, **P-Blue** cannot



Metrics: Normalized Joint Reward, Gini, Both Finished, Both Beat Baseline (BBB), Defection. Six models in self-play on all 80 boards.

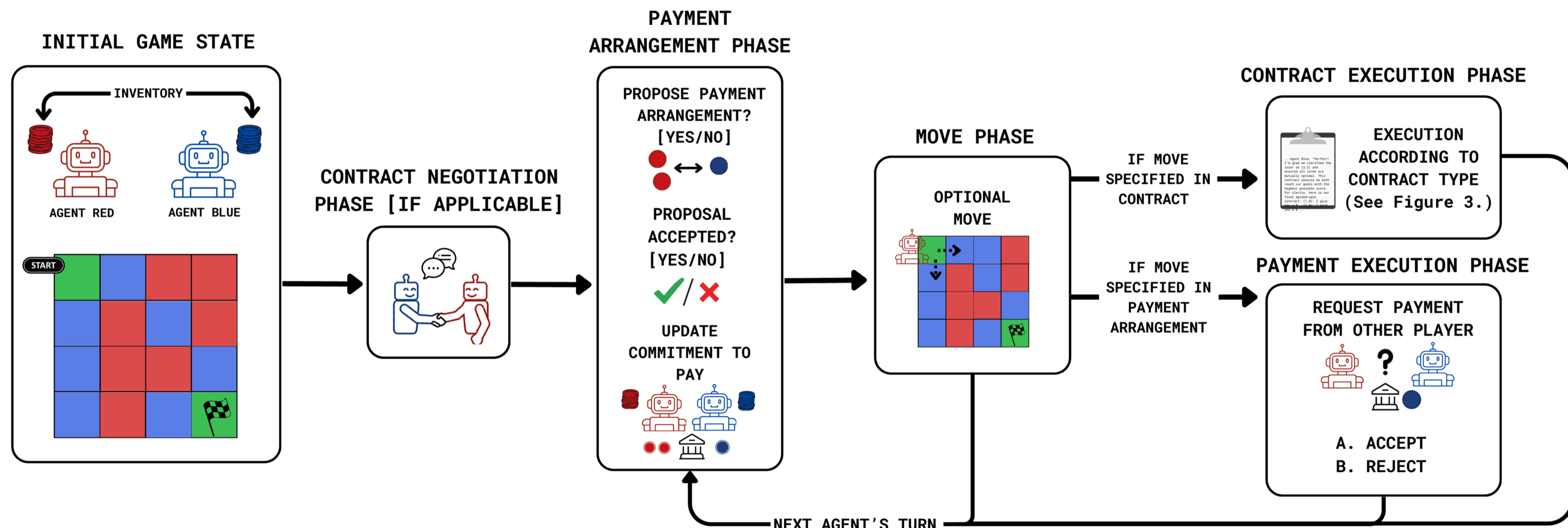
Without Contracts, Cooperation Fails Under Asymmetry

Pay-for-Partner, no contracts. $\hat{R}$ = Normalized Joint Reward; BF = Both Finished; Def. = Defection.

Model	R	Gini	BF	BBB	Def.
GPT-4.1	0.72	0.28	0.43	0.00	0.23
Haiku-4.5	0.91	0.10	0.82	0.00	0.01
LLaMA Maverick	0.97	0.05	0.95	0.00	0.13
LLaMA Scout	0.84	0.15	0.71	0.00	0.53
Qwen-3-235B	0.66	0.23	0.44	0.00	0.39
Qwen-3-30B	0.52	0.26	0.26	0.00	0.94
Mean	0.77	0.18	0.60	0.00	0.32

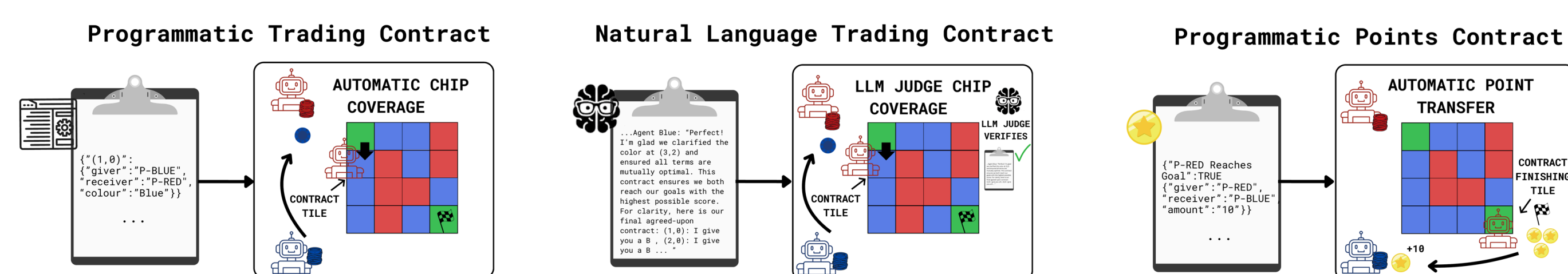
BBB = 0.00 for every model: no model got both players above baseline on any Asymmetric board.

The CT-Bench Environment



P-Red and **P-Blue** hold chips of only their own color; moving costs a chip of the tile's color, so finishing (20 points + 5 per chip) requires trades; Pay-for-Partner promises are **non-binding**.

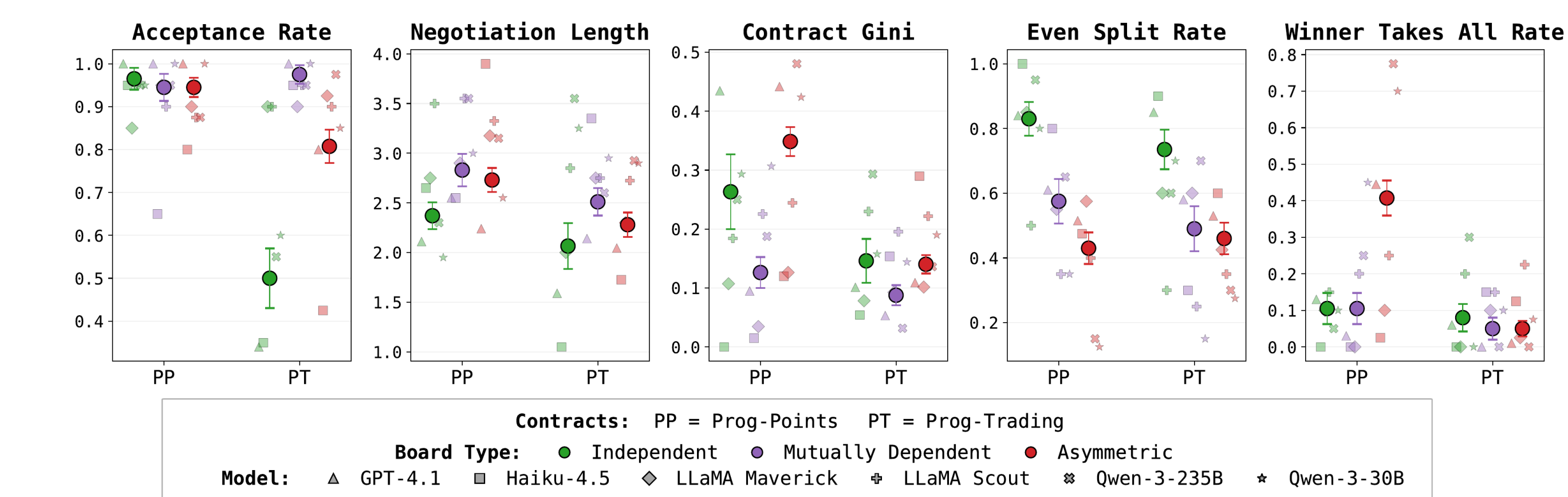
Self-Negotiated Contracts



Agents negotiate their own contracts:

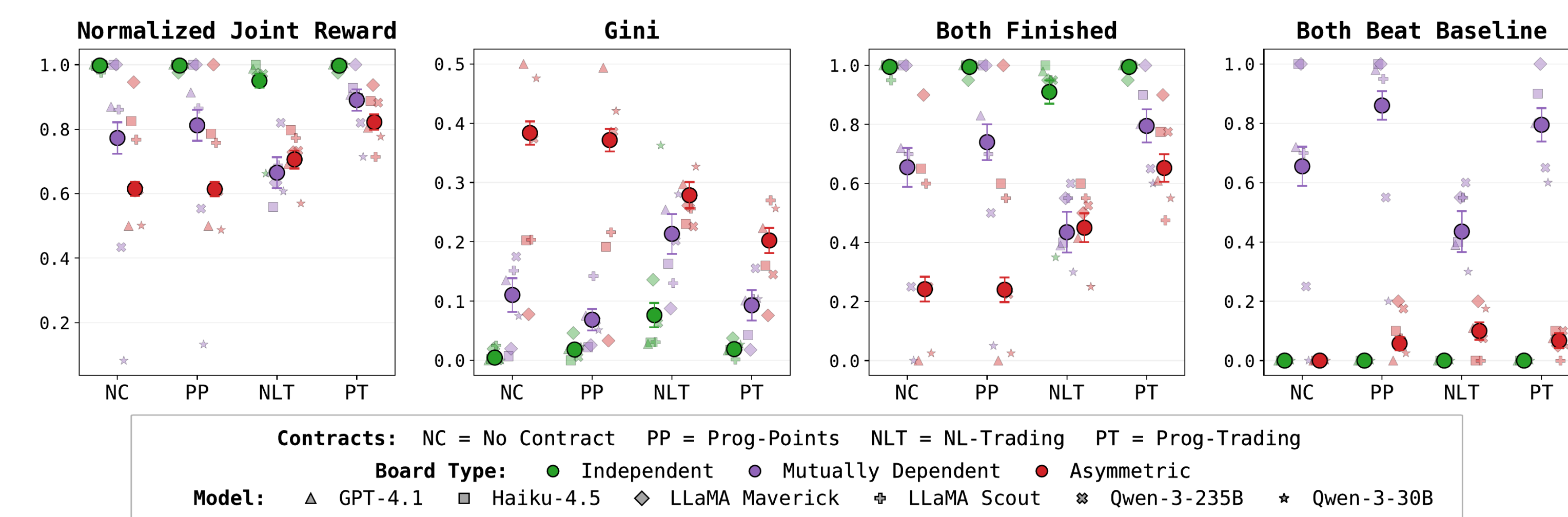
- Programmatic Trading**: chip transfers to cover agreed tiles, auto-executed
- Natural Language Trading**: contract in natural language, an LLM judge enforces it
- Programmatic Points**: promise part of your final score, auto-paid

Contract Formation



- Agents reliably agree: **86%** mean acceptance (Haiku-4.5: 67%)
- Representation shapes fairness**: Programmatic Points is winner-takes-all in 39%; Programmatic Trading splits equally in 49%

Contracts Improve Cooperative Outcomes



0.77 → 0.89

Joint Reward, No-Contract → Programmatic Trading ($p < 0.001$)

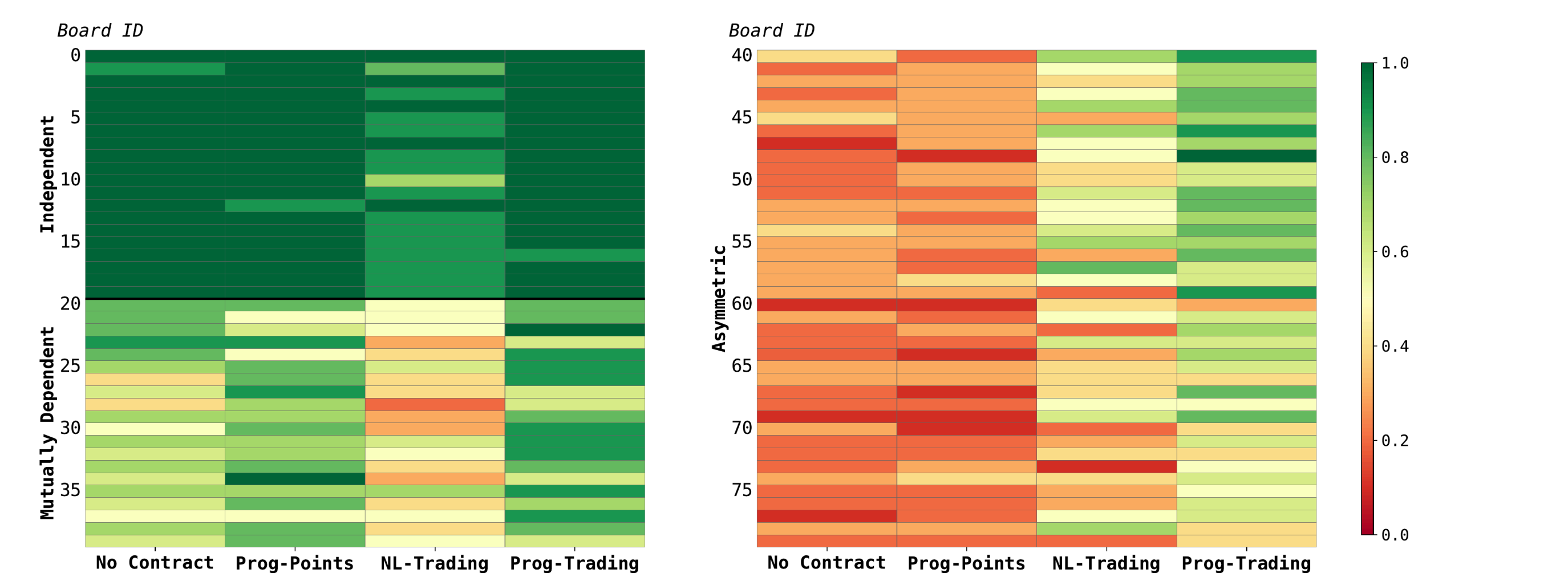
-38%

fewer non-binding promise arrangements

45 vs 6

of 60 dependent boards: best Both Finished

Both Finished, Board by Board



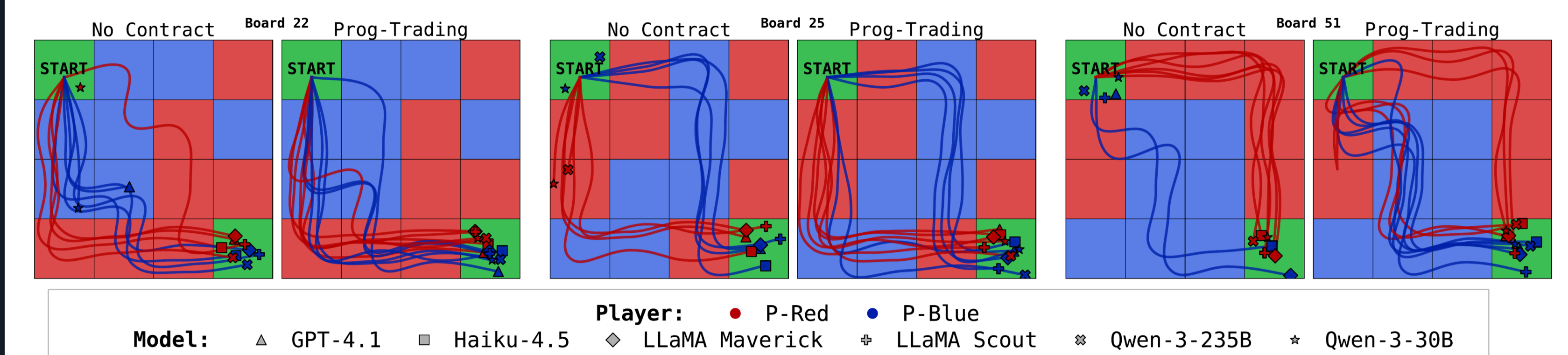
Programmatic Trading gives consistent gains on dependent boards.

When Contracts Backfire

On Mutually Dependent boards, natural-language contracts **underperform no contract at all**: Both Finished 46% vs 61% (No-Contract) vs 79% (Programmatic Trading).

Over-anchoring: contracts formed are similar, but Natural Language Trading suppresses the follow-up trading still needed (1.4 vs 3.0 proposals per game); models declining to trade cite the contract 94% vs 74% of the time.

Contracts Change Behavior on the Board



Contracts can improve outcomes. On boards where models fail to reach their goals without a contract, adding a Programmatic Trading contract lets every model complete its route.

Contracts Help the Weaker Player

On Asymmetric boards, **P-Blue's** reward is **68% higher** under Programmatic Trading ($p < 0.001$); **P-Red** gains nothing — the stronger player fails to exploit its position.

Beyond Self-Play

Pairing strong with weak models on Mutually Dependent boards: under Programmatic Trading the strong model lifts the weak one at its own expense — **cooperation survives large capability gaps**. Without contracts, GPT-4.1 scores 0 against Qwen-3-30B, which honors only **6%** of its promises (Haiku-4.5: 100%).

Takeaways

- The way contracts are represented and enforced matters**: contracts that compile to programs consistently outperform natural-language ones
- Self-negotiated contracts can improve cooperative outcomes** beyond regular trading
- Agents struggle to reach mutually beneficial outcomes from asymmetric starting points**

Paper & Acknowledgments



CT-Bench is open-sourced.
Paper: arxiv.org/abs/2607.22750
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